

Schematic configuration of the eight-state antiporter ordered binding transport mechanism

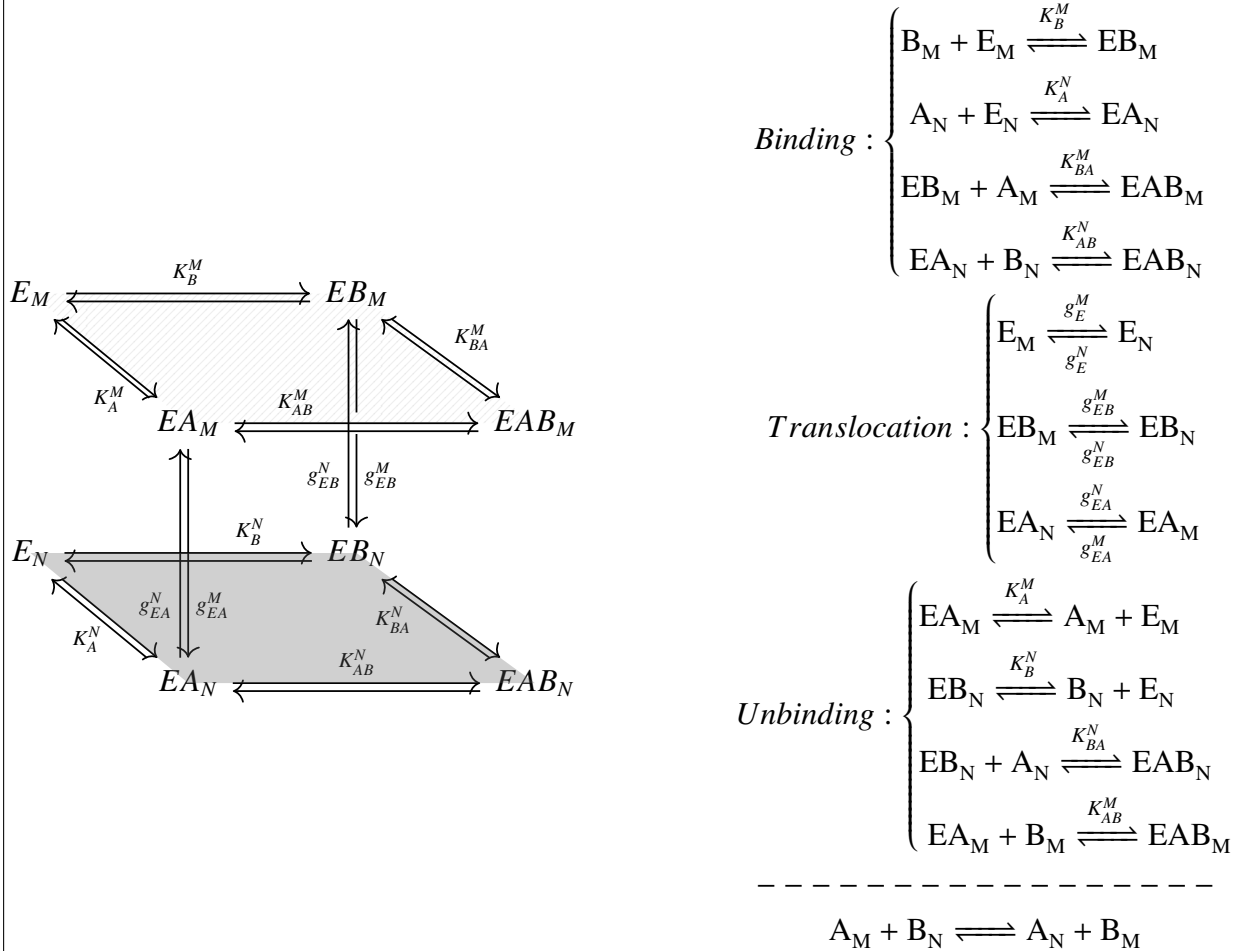


Figure 19: A schematic configuration of the antiporter **eight-state** transport mechanism and the resulting net transport flux equations. In developing this model, it is assumed that the dissociation constants for **binding reactions are not** affected by binding of A and B (which means to have $K_{AB}^{MorN} = K_B^{MorN}$ and $K_{BA}^{MorN} = K_A^{MorN}$). It is also assumed that: 1) the carrier is non-symmetric and 2) binding reaction occur much faster relative to the translocation of the carrier (rapid binding assumption) (part 1/2 continued on the next page).

Net Transport Fluxes

$$J_{antiporter}^{M,N(net)} = [E]_t \frac{(g_{EA}^M \alpha^M)(g_{EB}^N \beta^N) - (g_{EA}^N \alpha^N)(g_{EB}^M \beta^M)}{R_M R_{NN} + R_N R_{MM}} \quad (62a)$$

$$J_{A,antiporter}^{M,N(net)} = [E]_t \frac{(g_{EA}^M \alpha^M)(g_{EB}^N \beta^N) - (g_{EA}^N \alpha^N)(g_{EB}^M \beta^M)}{R_M R_{NN} + R_N R_{MM}} \quad (62b)$$

$$J_{B,antiporter}^{M,N(net)} = [E]_t \frac{(g_{EB}^M \beta^M)(g_{EA}^N \alpha^N) - (g_{EB}^N \beta^N)(g_{EA}^M \alpha^M)}{R_M R_{NN} + R_N R_{MM}} \quad (62c)$$

where

$$[E]_t = [E]_M + [EA]_M + [EB]_M + [EAB]_M + [EAB]_N + [EB]_N + [EA]_N + [E]_N$$

$$\alpha^M = \frac{[A]_M}{K_A^M}, \quad \beta^M = \frac{[B]_M}{K_B^M}$$

$$R_M = 1 + \alpha^M + \beta^M + \alpha^M \beta^M$$

$$R_{MM} = g_{EA}^M \alpha^M + g_{EB}^M \beta^M$$

$$\alpha^N = \frac{[A]_N}{K_A^N}, \quad \beta^N = \frac{[B]_N}{K_B^N}$$

$$R_N = 1 + \alpha^N + \beta^N + \alpha^N \beta^N$$

$$R_{NN} = g_{EA}^N \alpha^N + g_{EB}^N \beta^N$$

Figure 19: A schematic configuration of the antiporter eight-state transport mechanism and the resulting net transport flux equations. In developing this model, it is assumed that the dissociation constants for binding reactions are not affected by binding of A and B (which means to have $K_{AB}^{MorN} = K_B^{MorN}$ and $K_{BA}^{MorN} = K_A^{MorN}$). It is also assumed that: 1) the carrier is non-symmetric and 2) binding reaction occur much faster relative to the translocation of the carrier (rapid binding assumption) (part 2/2 continued from previous page).

Six-state antiporter ordered binding model:

The six-state model is very similar to the eight-state model, except that A and B cannot bind simultaneously to the transporter, and consequently, the formation of the complexes EAB_M and EAB_N does not occur. Figure 20 shows the "ping-pong" model of a six-state antiport transport model. As shown in Figure 20, in the first step, component A binds to the protein on the outside surface. Assuming that the binding of solute B to the carrier occurs simultaneously on the other side of the membrane, the A-carrier (EA_M) and B-carrier (EB_N) complexes are formed on opposite sides of the membrane. After translocation of the formed complexes, each of the complexes undergoes a dissociation reaction. These dissociation reactions lead to the release of solutes A and B from the N (inside) and M (outside) faces of the membrane, respectively. The A and B fluxes driven from region M to N via this antiport mechanism must be equal and opposite (i.e., $J_{B,antiporter}^{M,N} = -J_{A,antiporter}^{M,N}$). The six-state model was first developed by Weinstein et al. (2000)